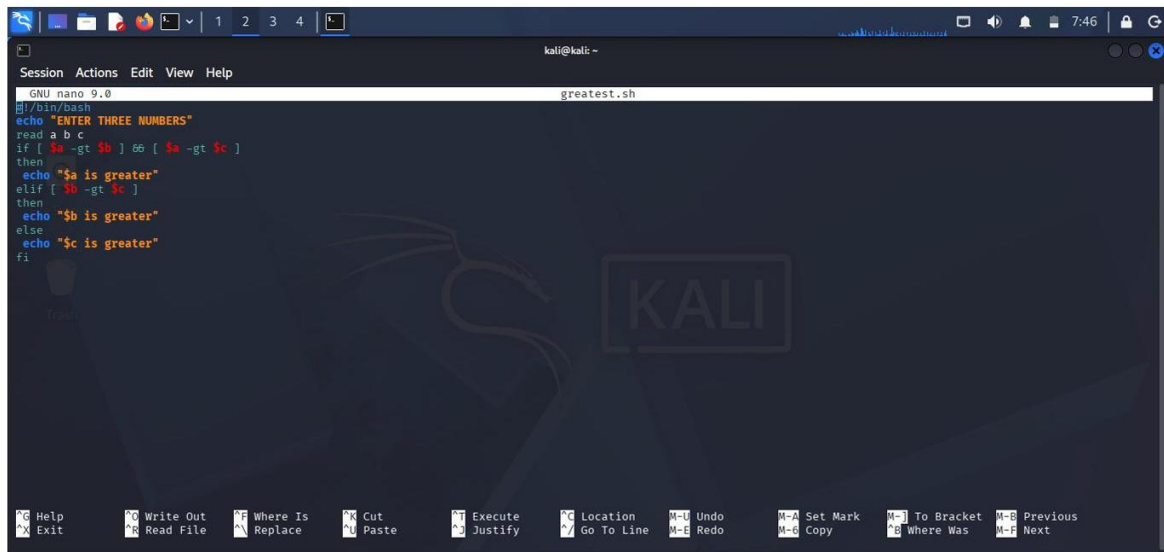


SHELL PROGRAMMING :GREATEST AMONG THREENUMBERS:

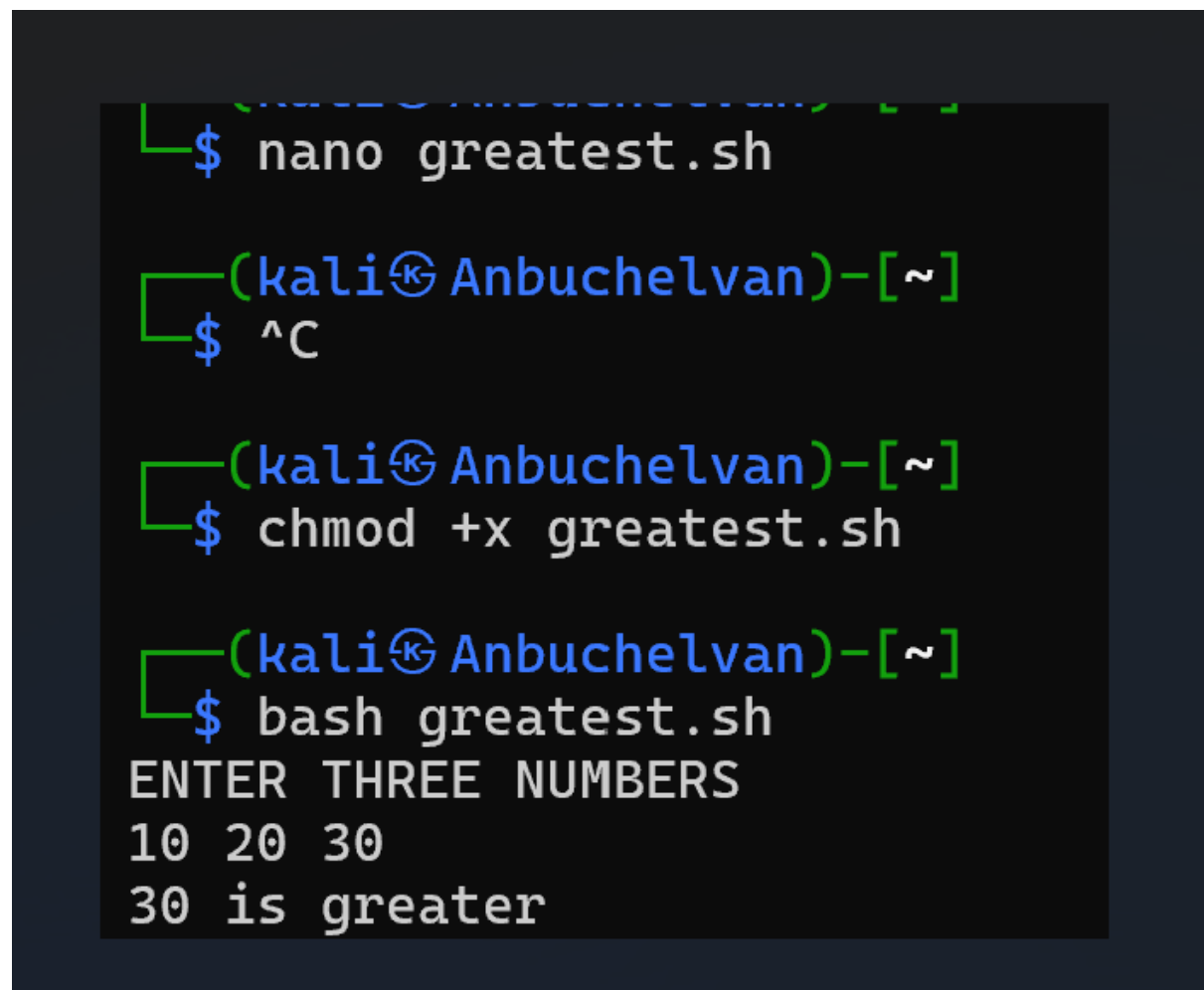


The screenshot shows a terminal window with the title 'kali@kali: ~'. The user is in the nano text editor editing a file named 'greatest.sh'. The script content is as follows:

```
#!/bin/bash
echo "ENTER THREE NUMBERS"
read a b c
if [ $a -gt $b ] && [ $a -gt $c ]
then
echo "$a is greater"
elif [ $b -gt $c ]
then
echo "$b is greater"
else
echo "$c is greater"
fi
```

The terminal window includes a menu bar with options: Session, Actions, Edit, View, Help. At the bottom, there is a toolbar with various icons and keyboard shortcuts like Ctrl+H (Help), Ctrl+X (Exit), Ctrl+W (Write Out), Ctrl+R (Read File), Ctrl+F (Where Is), Ctrl+M (Replace), Ctrl+K (Cut), Ctrl+V (Paste), Ctrl+J (Execute), Ctrl+I (Justify), Ctrl+G (Location), Ctrl+T (Go To Line), Ctrl+Z (Undo), Ctrl+Y (Redo), Ctrl+M (Set Mark), Ctrl+C (Copy), Ctrl+B (To Bracket), Ctrl+W (Where Was), Ctrl+P (Previous), and Ctrl+N (Next).

OUTPUT:

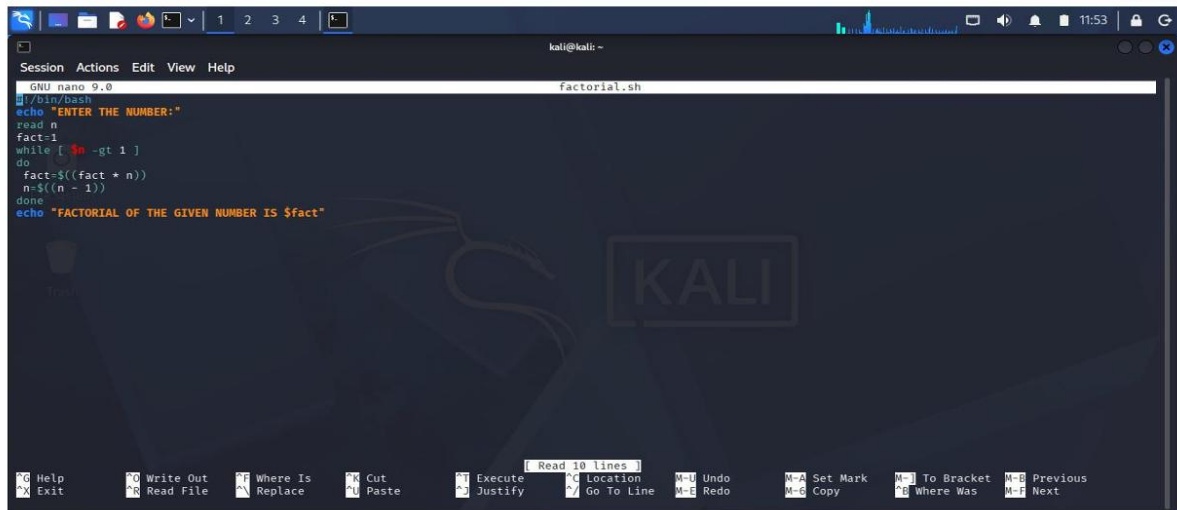


The screenshot shows a terminal window with the title 'kali@kali: ~'. The user is in the nano text editor editing a file named 'greatest.sh'. The script content is as follows:

```
#!/bin/bash
echo "ENTER THREE NUMBERS"
read a b c
if [ $a -gt $b ] && [ $a -gt $c ]
then
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elif [ $b -gt $c ]
then
echo "$b is greater"
else
echo "$c is greater"
fi
```

The terminal window includes a menu bar with options: Session, Actions, Edit, View, Help. At the bottom, there is a toolbar with various icons and keyboard shortcuts like Ctrl+H (Help), Ctrl+X (Exit), Ctrl+W (Write Out), Ctrl+R (Read File), Ctrl+F (Where Is), Ctrl+M (Replace), Ctrl+K (Cut), Ctrl+V (Paste), Ctrl+J (Execute), Ctrl+I (Justify), Ctrl+G (Location), Ctrl+T (Go To Line), Ctrl+Z (Undo), Ctrl+Y (Redo), Ctrl+M (Set Mark), Ctrl+C (Copy), Ctrl+B (To Bracket), Ctrl+W (Where Was), Ctrl+P (Previous), and Ctrl+N (Next).

FACTORIAL OF A GIVEN NUMBER :

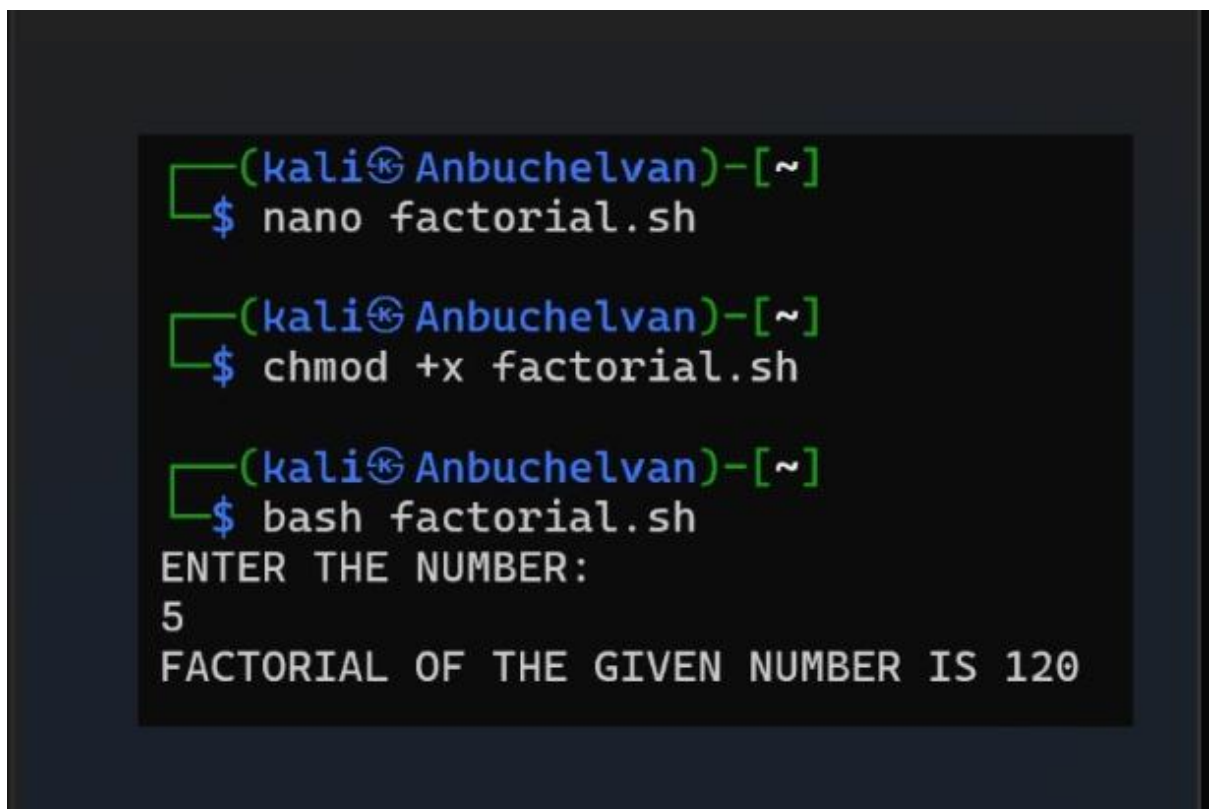


A screenshot of a terminal window on a Kali Linux system. The window title is 'kali@kali: ~'. The terminal shows the nano 9.0 editor editing a file named 'factorial.sh'. The script content is as follows:

```
#!/bin/bash
echo "ENTER THE NUMBER:"
read n
fact=1
while [ $n -gt 1 ]
do
    fact=$((fact * n))
    n=$((n - 1))
done
echo "FACTORIAL OF THE GIVEN NUMBER IS $fact"
```

The nano editor interface includes a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. At the bottom, there is a toolbar with various shortcuts like 'Ctrl+G Help', 'Ctrl+O Write Out', etc.

OUTPUT:



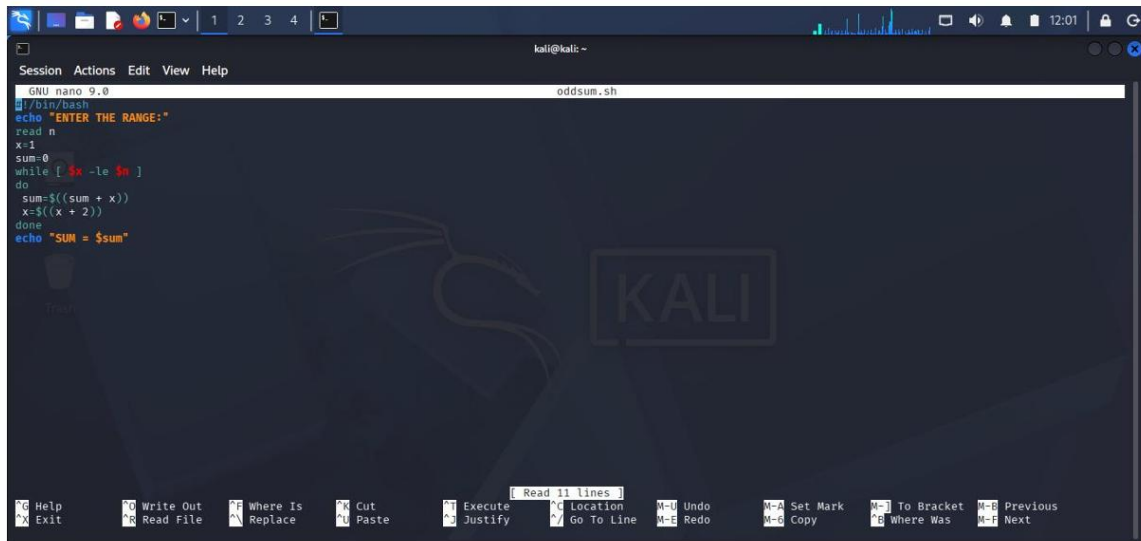
A screenshot of a terminal window showing the execution of the 'factorial.sh' script. The prompt is '(kaliⓀ Anbuchelvan)-[~]'. The user enters the following commands:

```
$ nano factorial.sh
$ chmod +x factorial.sh
$ bash factorial.sh
```

The output of the script is:

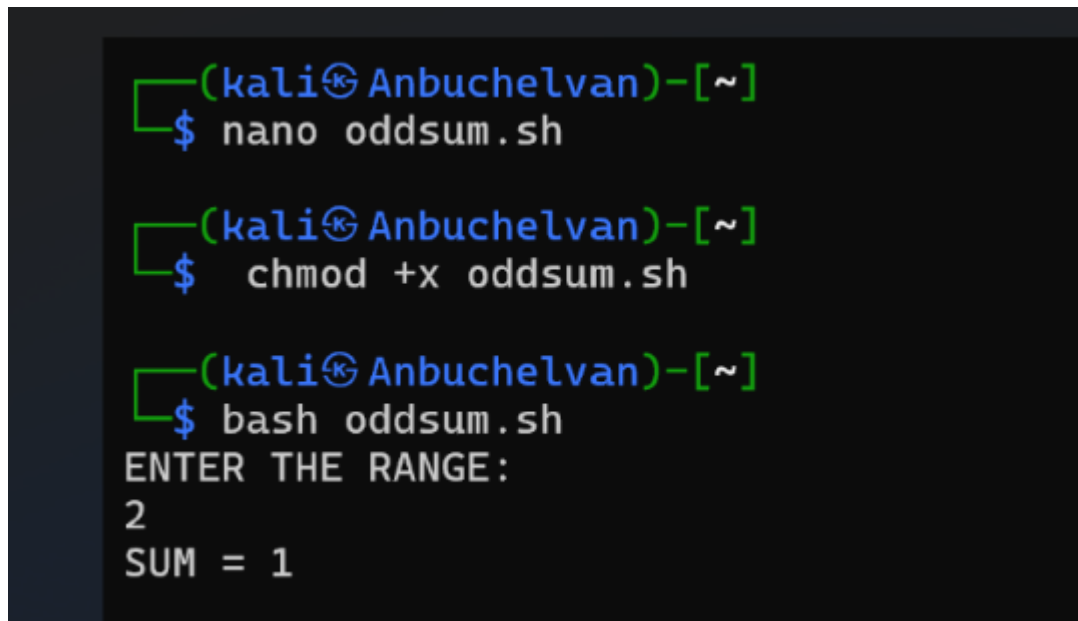
```
ENTER THE NUMBER:
5
FACTORIAL OF THE GIVEN NUMBER IS 120
```

SUM OF ODD NUMBERS:



```
GNU nano 9.0 oddsum.sh
echo "ENTER THE RANGE:"
read n
x=1
sum=0
while [ $x -le $n ]
do
sum=$((sum + x))
x=$((x + 2))
done
echo "SUM = $sum"
```

OUTPUT:

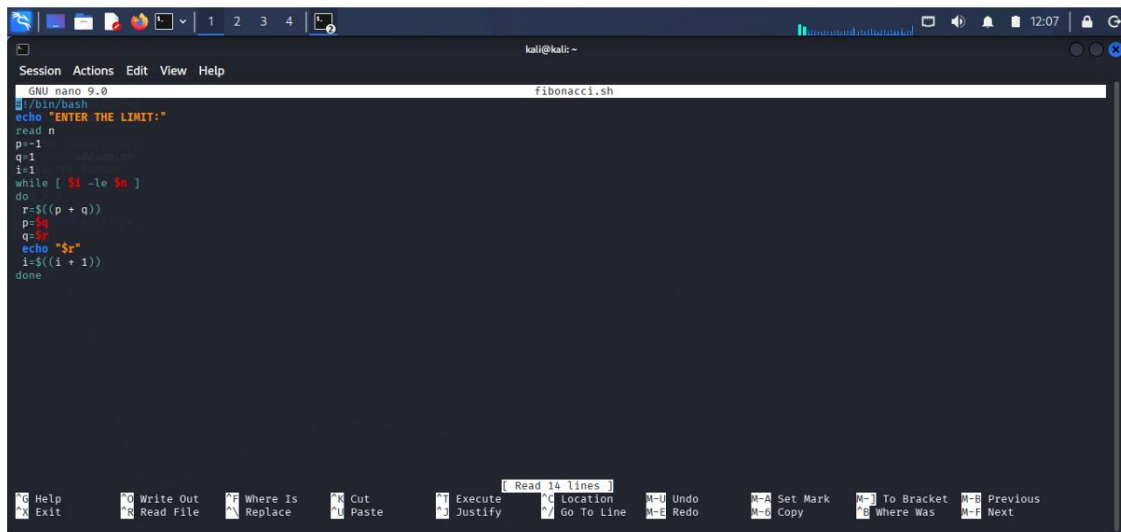


```
(kali@Anbuchelvan)~$ nano oddsum.sh

(kali@Anbuchelvan)~$ chmod +x oddsum.sh

(kali@Anbuchelvan)~$ bash oddsum.sh
ENTER THE RANGE:
2
SUM = 1
```

FIBONACCI NUMBER:

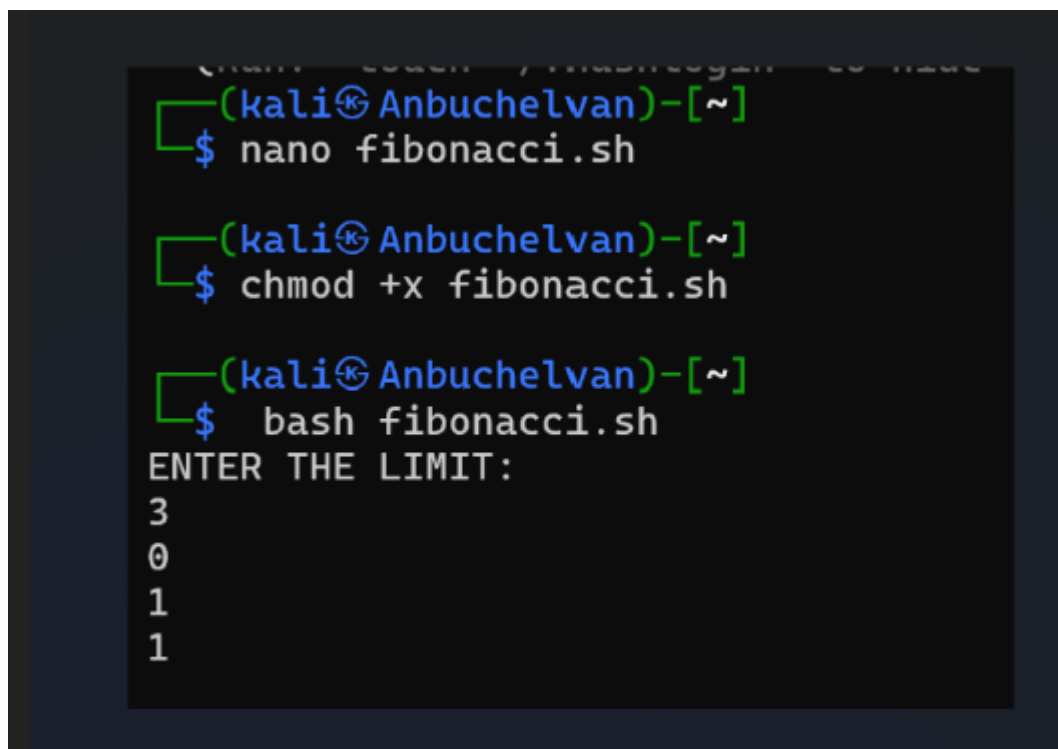


A screenshot of a terminal window with a dark background. The window title is 'kali@kali: ~'. The terminal shows the nano 9.0 editor editing a file named 'fibonacci.sh'. The script content is as follows:

```
#!/bin/bash
echo "ENTER THE LIMIT:"
read n
p=1
q=1
i=1
while [ $i -le $n ]
do
r=$((p + q))
p=$q
q=$r
echo "$r"
i=$((i + 1))
done
```

The bottom of the window shows a menu bar with various options: Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, Next.

OUTPUT:



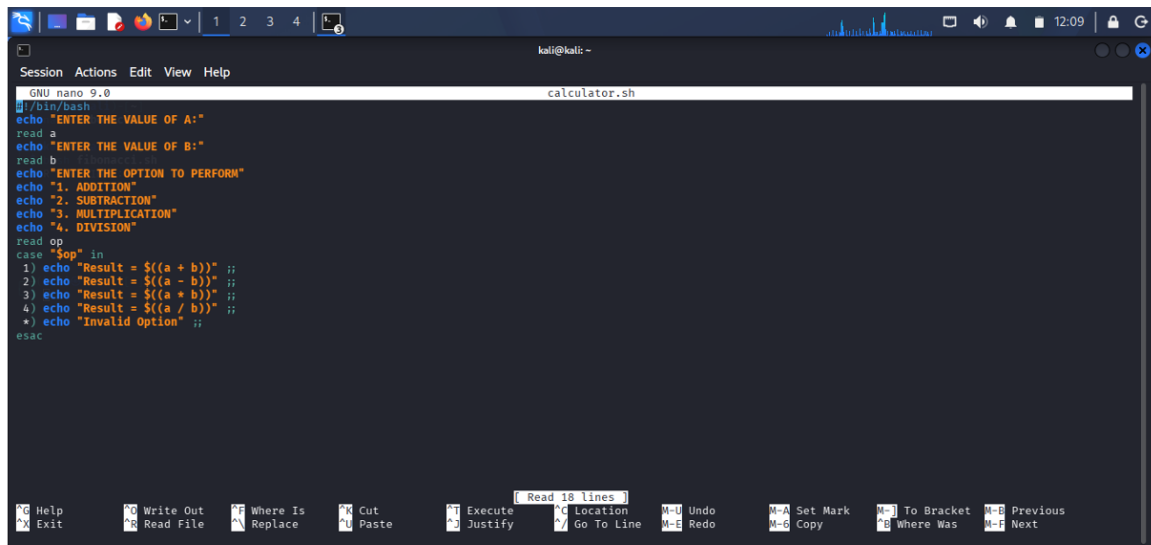
A screenshot of a terminal window showing the execution of the 'fibonacci.sh' script. The prompt is '(kali@Anbuchelvan)~'. The user enters 'nano fibonacci.sh', then 'chmod +x fibonacci.sh', and finally 'bash fibonacci.sh'. The script prompts 'ENTER THE LIMIT:' and the user enters '3'. The script then outputs the Fibonacci sequence: '0', '1', '1'.

```
(kali@Anbuchelvan)~
$ nano fibonacci.sh

(kali@Anbuchelvan)~
$ chmod +x fibonacci.sh

(kali@Anbuchelvan)~
$ bash fibonacci.sh
ENTER THE LIMIT:
3
0
1
1
```

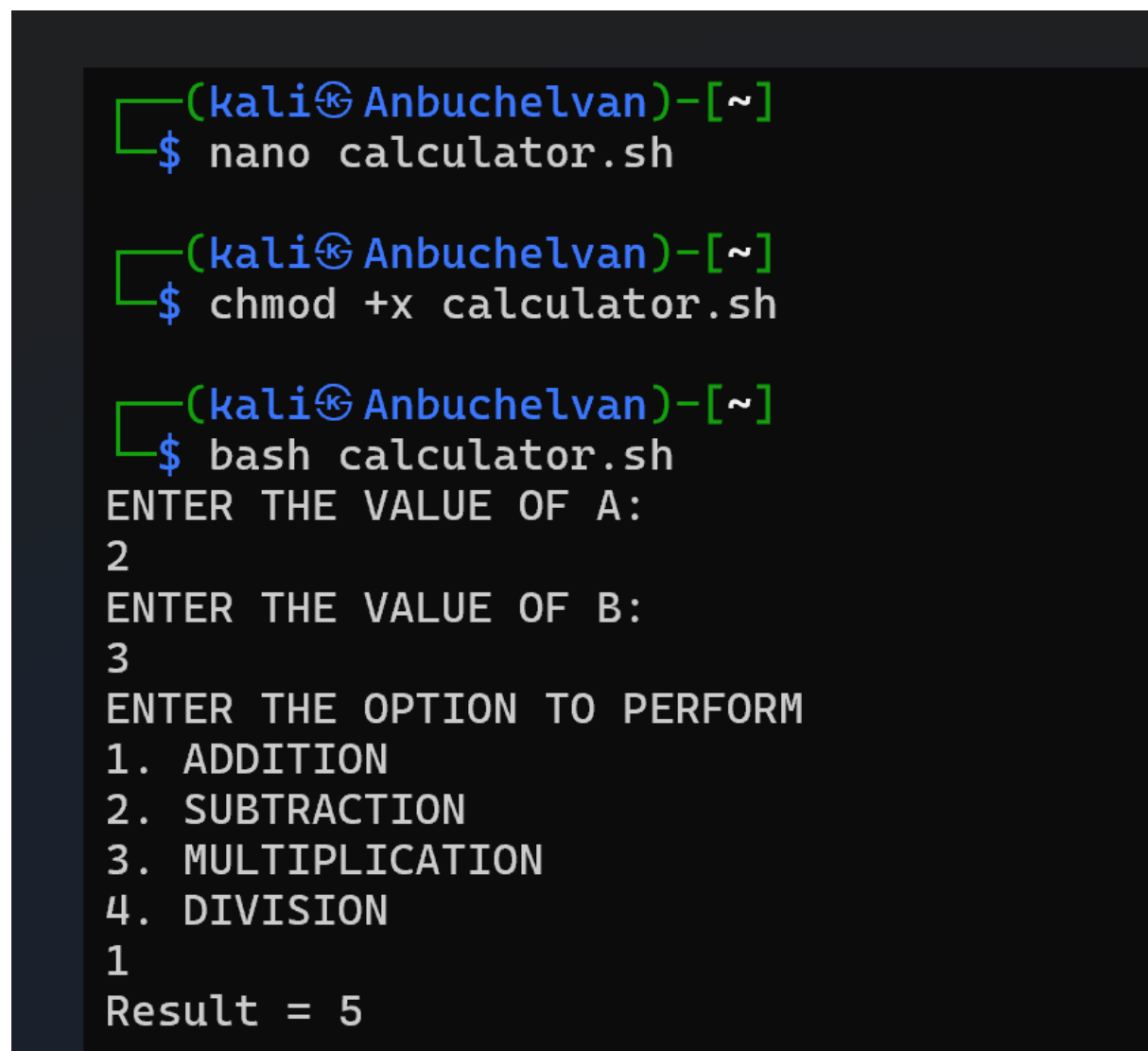
ARITHMETIC CALCULATOR:



The screenshot shows a terminal window with the nano text editor open. The script 'calculator.sh' is being edited. It prompts the user to enter values for A and B, and then to select an operation from a menu. The script uses a case statement to perform the selected operation. The terminal window has a title bar with 'kali@kali: ~' and a status bar with various keyboard shortcuts.

```
GNU nano 9.0 calculator.sh
#!/bin/bash
echo "ENTER THE VALUE OF A:"
read a
echo "ENTER THE VALUE OF B:"
read b
echo "ENTER THE OPTION TO PERFORM"
echo "1. ADDITION"
echo "2. SUBTRACTION"
echo "3. MULTIPLICATION"
echo "4. DIVISION"
read op
case "$op" in
1) echo "Result = $((a + b))" ;;
2) echo "Result = $((a - b))" ;;
3) echo "Result = $((a * b))" ;;
4) echo "Result = $((a / b))" ;;
*) echo "Invalid Option" ;;
esac
```

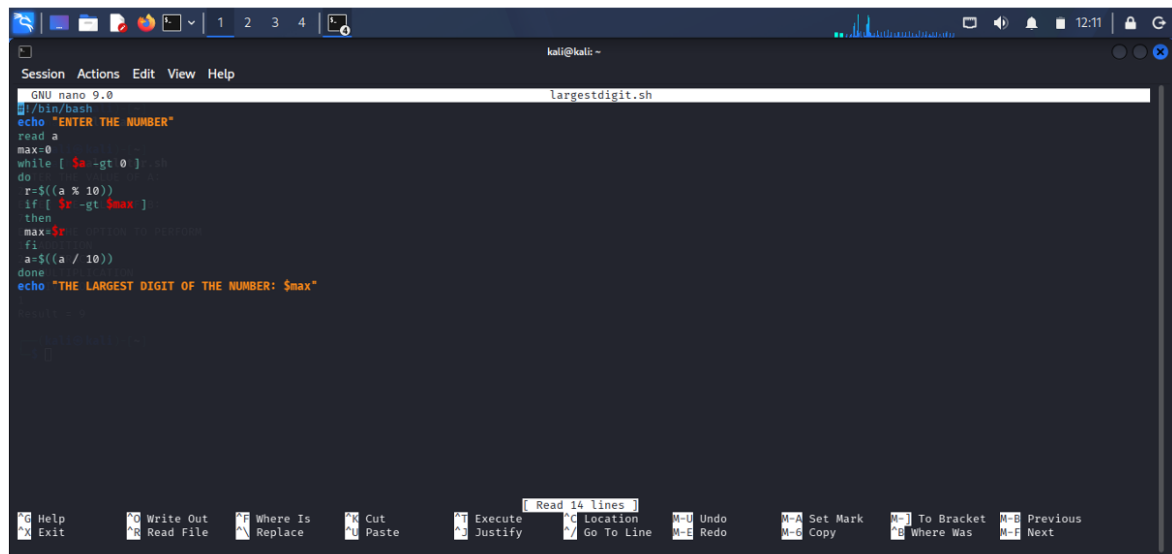
OUTPUT:



The screenshot shows the output of the calculator script. The user has entered 2 for A, 3 for B, and 1 for the operation (Addition). The script has calculated the result as 5.

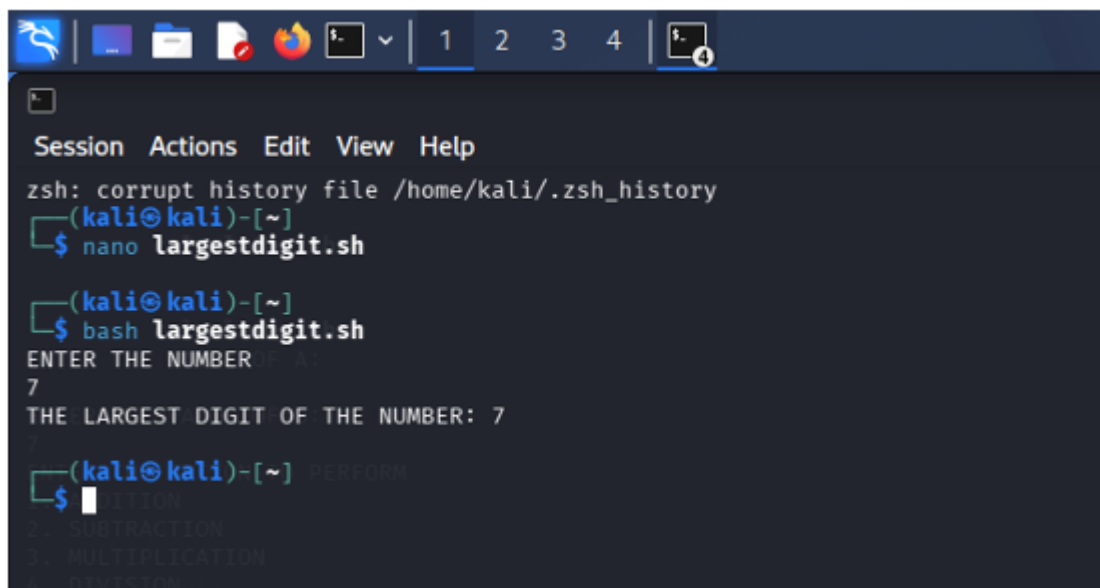
```
(kali@Anbuchelvan)~$ nano calculator.sh
(kali@Anbuchelvan)~$ chmod +x calculator.sh
(kali@Anbuchelvan)~$ bash calculator.sh
ENTER THE VALUE OF A:
2
ENTER THE VALUE OF B:
3
ENTER THE OPTION TO PERFORM
1. ADDITION
2. SUBTRACTION
3. MULTIPLICATION
4. DIVISION
1
Result = 5
```

LARGEST DIGIT OF A NUMBER:



```
GNU nano 9.0 largestdigit.sh
#!/bin/bash
echo "ENTER THE NUMBER"
read a
max=0
while [ $a -gt 0 ]
do
r=$((a % 10))
if [ $r -gt $max ]
then
max=$r
fi
a=$((a / 10))
done
echo "THE LARGEST DIGIT OF THE NUMBER: $max"
```

OUTPUT:

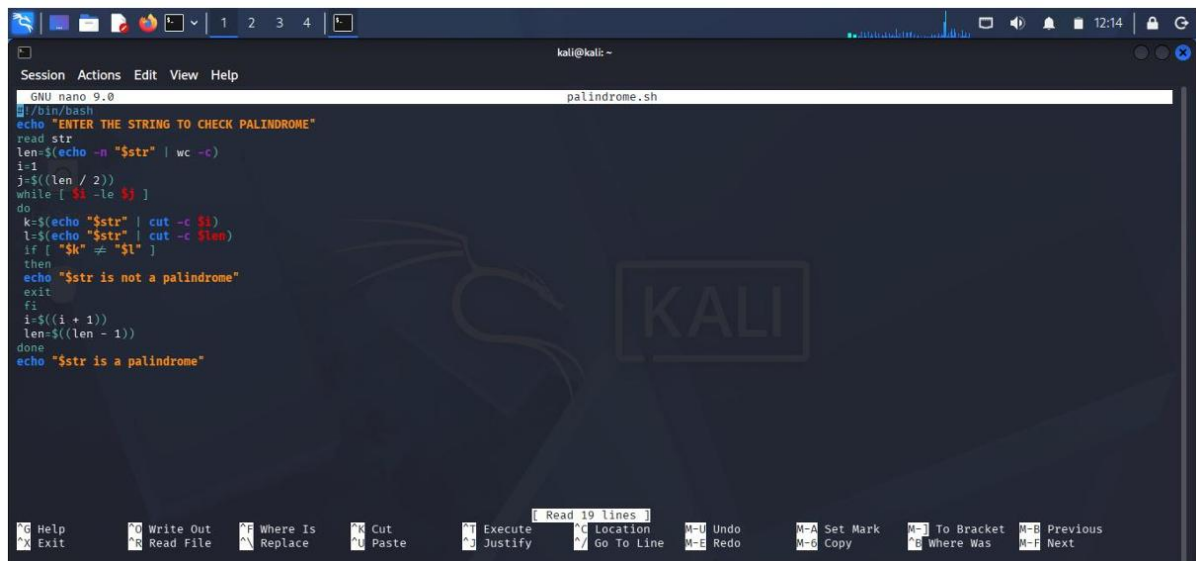


```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ nano largestdigit.sh

(kali@kali)-[~]
$ bash largestdigit.sh
ENTER THE NUMBER: 7
THE LARGEST DIGIT OF THE NUMBER: 7

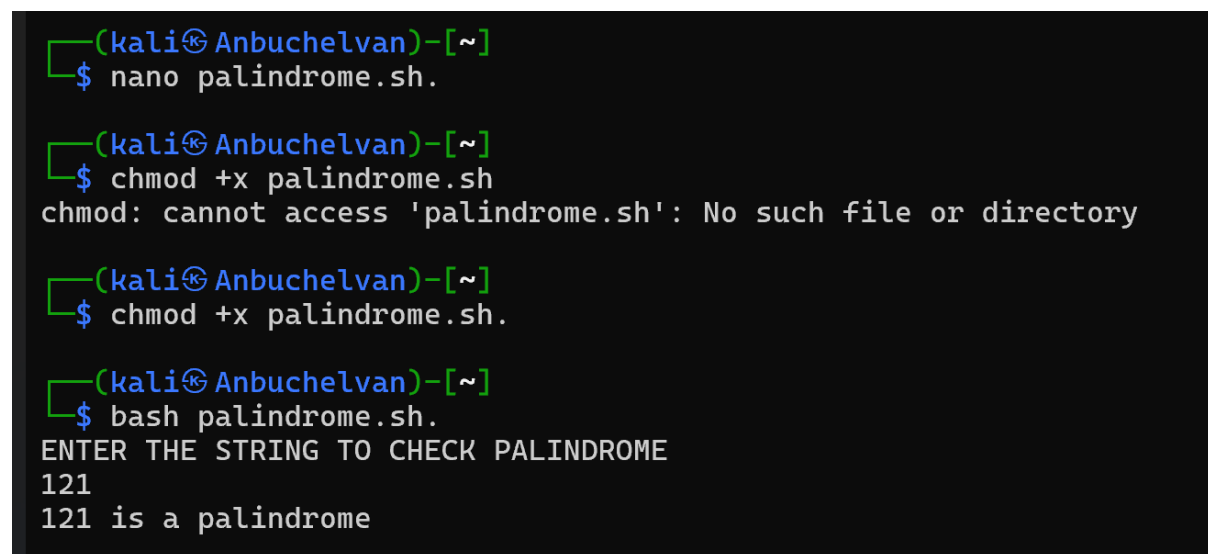
(kali@kali)-[~]
$
```

PALINDROME OR NOT:



```
GNU nano 9.0 palindrome.sh
#!/bin/bash
echo "ENTER THE STRING TO CHECK PALINDROME"
read str
len=$(echo -n "$str" | wc -c)
i=1
j=$((len / 2))
while [ $i -le $j ]
do
k=$(echo "$str" | cut -c $i)
l=$(echo "$str" | cut -c $len)
if [ "$k" != "$l" ]
then
echo "$str is not a palindrome"
exit
fi
i=$((i + 1))
len=$((len - 1))
done
echo "$str is a palindrome"
```

OUTPUT:



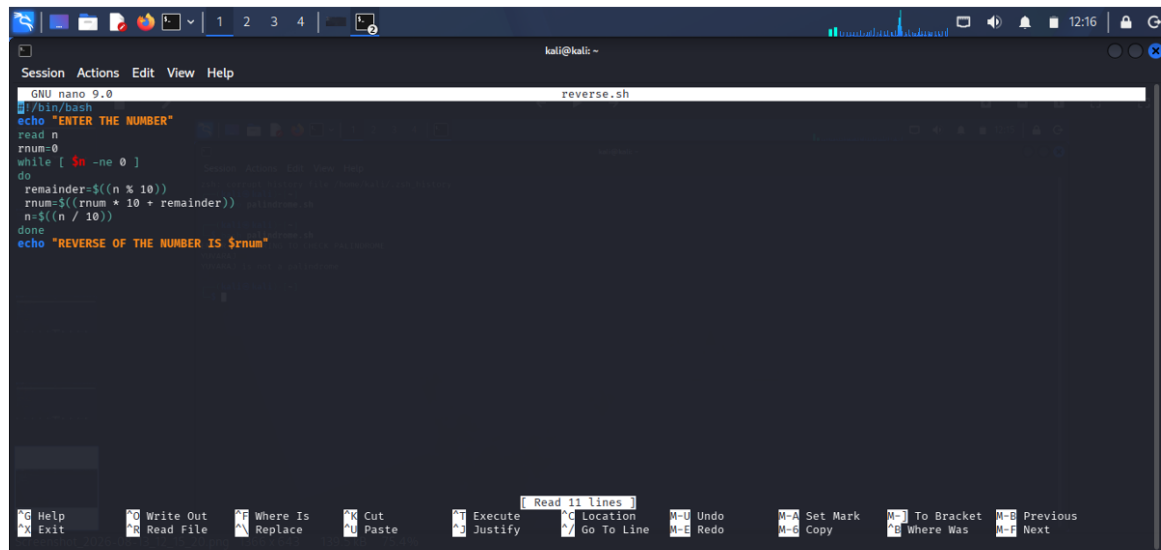
```
(kali@Anbuchelvan)-[~]
$ nano palindrome.sh.

(kali@Anbuchelvan)-[~]
$ chmod +x palindrome.sh
chmod: cannot access 'palindrome.sh': No such file or directory

(kali@Anbuchelvan)-[~]
$ chmod +x palindrome.sh.

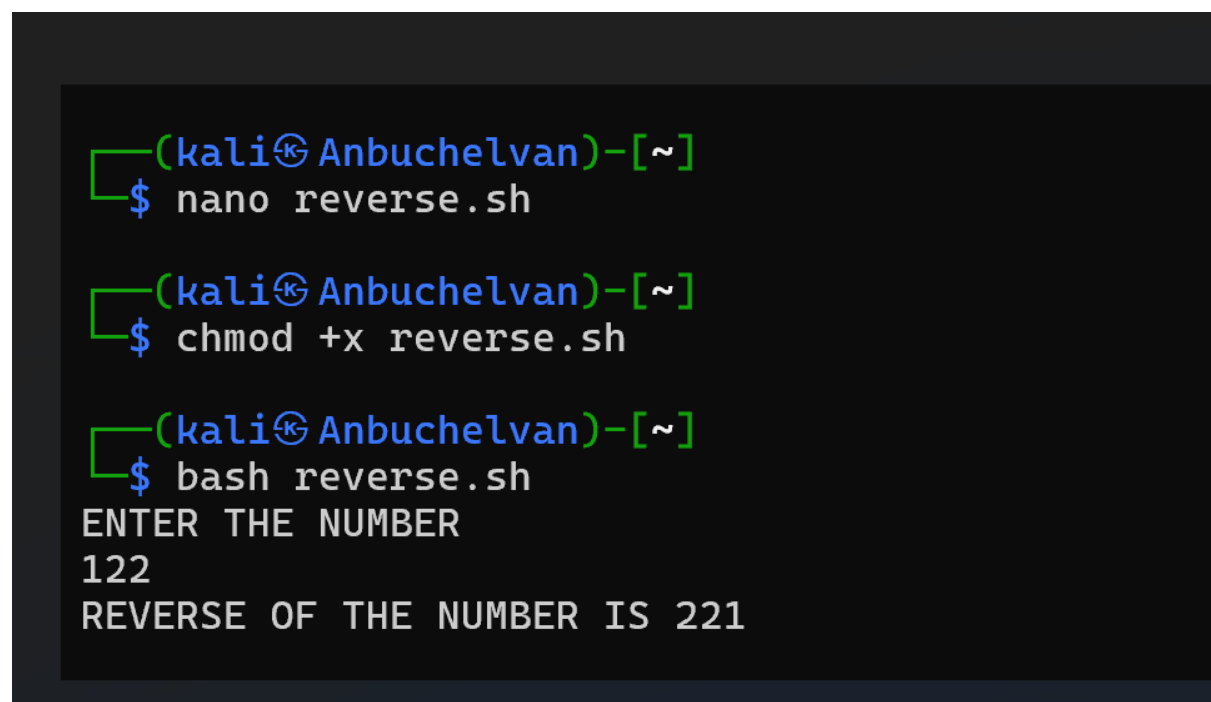
(kali@Anbuchelvan)-[~]
$ bash palindrome.sh.
ENTER THE STRING TO CHECK PALINDROME
121
121 is a palindrome
```

REVERSE OF A GIVEN NUMBER:



```
GNU nano 9.0 reverse.sh
#!/bin/bash
echo "ENTER THE NUMBER"
read n
rnum=0
while [ $n -ne 0 ]
do
    remainder=$((n % 10))
    rnum=$((rnum * 10 + remainder))
    n=$((n / 10))
done
echo "REVERSE OF THE NUMBER IS $rnum"
```

OUTPUT:



```
(kaliⓀ Anbuchelvan)-[~]
$ nano reverse.sh

(kaliⓀ Anbuchelvan)-[~]
$ chmod +x reverse.sh

(kaliⓀ Anbuchelvan)-[~]
$ bash reverse.sh
ENTER THE NUMBER
122
REVERSE OF THE NUMBER IS 221
```